Problem 1

Task:

1. Close-loop Transfer function can be found either using block diagram simplification. For unity feedback system in Problem 1 with tranfsre function $G = \frac{200}{s^4+6s^3+11s^2+6s}$ the closed loop transfer function G_{cl} is:

$$G_{cl} = \frac{G}{1+G} = \frac{200}{s^4+6s^3+11s^2+6s+200} \tag{1}$$

   or Matlab command feedback(G,1) can also be used. Ref to m-file solution for details.

2. The system has 4 poles and no zeros. See m-file solution for details..

3. Two poles are in right half plane thus system is unstable.

4. Step response is unstable as expected due to poles in right half plane.

Problem 2

1. The stability of a closed loop system can be found by drawing root locus of the loop transfer function. Generate root locus for a unity feedback system using Matlab command rlocus(K*G), where $K = 1$. Find the gain where locus crosses to right half plane. The gain at which the locus crosses imaginary axis is 2, thus the range of gain for which the system is stable is $K < 2$;.

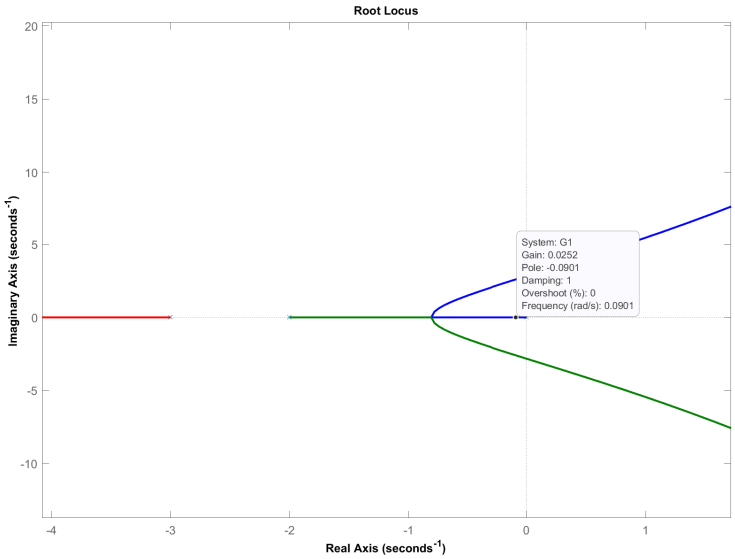


Figure 1: Root locus plot

Problem 3

1. Eigen values can be found by calculating the roots of the determinant:

$$|sI - A| = 0 \tag{2}$$

use Matlab command eig(A). Ref . to Matlab file for details.

2. The system has one real pole on the left half-plane and two complex poles in the right half-plane, the system is unstable. Details in Matlab solution.

Problem 4

1. The system is stable for $K = 20$ and unstable for $K = 200$.

2. To determine the stable range for gain K use root locus plot of loop transfer function $G(s)H(s)$ with $K = 1$. The root locus plot is shown in Figure 2. It can be seen that the

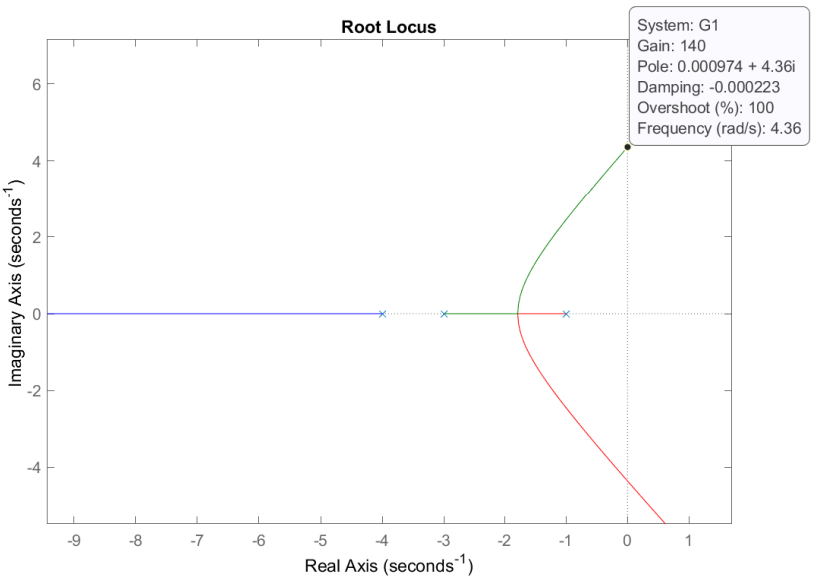


Figure 2:

system is stable for $K < 140$.

3. Frequency response of loop transfer function $G(s)H(s)$ with $K = 1$ is shown in Figure 3.

4. Gain margin G_M refers to the gain required to raise the magnitude curve to 0 dB at teh frequency where the phase of the system is 180°. In other words it is additional agin required to make the system unstable. Gain marginn is 42.9 dB.

$$\begin{aligned} G_M &= 42.9\text{dB} \\ &= 20\log_{10}(K) \implies K = 139.64 \end{aligned} \tag{3}$$

This confirms the finding from root locus that K¡139.64 will keep the system stable. As the magnitude curve does not cross 0 dB, thus phase margin is infinite (PM=∞).

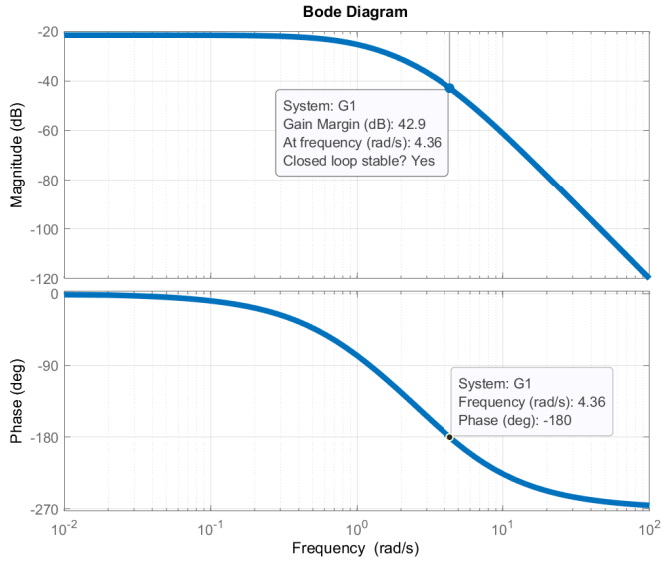


Figure 3:

Problem 5

1. Frequency response of the open-loop system with $K = 1$ is shown in Figure 1.

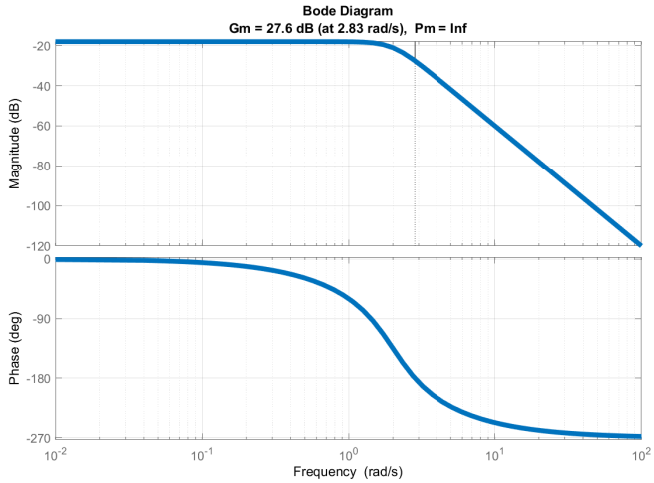


Figure 4:

2. Gain Margin (G_M) and Phase margin ϕ_M are

$$\begin{aligned} G_M &= 27.6\text{dB} \\ \omega_{GM} &= 2.83\text{rad/s} \\ \phi_M &= \infty \end{aligned} \tag{4}$$

3. It can be seen from gain margin that $K < 27.6$ dB will give a stable system. $K < 27.6\text{dB} = 10^{(27.6/20)} = 23.98$. Thus $K < 23.98$ will keep the system stable.